

Profile

Software Engineer skilled in C, Python, and AI/ML with experience in real-time and distributed system design.Strong foundation in computer vision, embedded systems, Linux-based development, and SDLC practices.Proven ability to build scalable, performance-driven systems with microservices and cloud technologies.Focused on applying AI/ML and software engineering to power systems, control, and industrial automation.

Education

Master Of Technology(Data Science and Artificial Intelligent), Indian Institute Of Information Technology Design and Manufacturing, Kancheepuram CGPA:8.67	08/2025 – Present Chennai,India
Bachelor of Technology(Computer science and Engineering), LBS COLLEGE OF ENGINEERING CGPA:8.85	07/2020 – 05/2024 Kasaragod,India
Graduate Aptitude Test in Engineering GATE Score:476 AIR:8357	02/2025

Projects

Clinical Decision Support System — Multi-Agent Distributed AI Platform • Architected a distributed 7-microservice system using Python and FastAPI with asynchronous communication for real-time workflows. • Designed parallel multi-agent execution using asyncio, reducing latency and improving system responsiveness. • Built scalable API-driven services (JSON-RPC, HTTP) ensuring modular and fault-tolerant design. • Implemented audit logging and applied AI/ML models with validation, fallback logic, and reliability constraints.	03/2026 – 04/2026
AI-Based Traffic Violation Detection System (Attention-based CV) • Developed a real-time traffic violation detection system using OpenCV and YOLO-based models for video stream analysis. • Designed an optimized inference pipeline to improve robustness under varying environmental conditions. • Implemented a low-latency workflow suitable for real-time and edge/embedded deployment.	09/2025
Real-Time Camera Disturbance & Monitoring System (Anomaly Detection) • Developed a real-time anomaly detection system using CV/ML to detect occlusion, glare, signal loss, and frame freezing. • Designed a multi-threaded pipeline for concurrent stream processing and efficient CPU utilization. • Built a modular system optimized for low-latency and deployment on resource-constrained environments.	09/2025 – 11/2025
Malayalam OCR System (CV + NLP + Sequence Modeling) • Developed an OCR system using CNN + CRNN with CTC decoding for sequence-based text recognition. • Applied image preprocessing and CV techniques for improved accuracy and feature extraction. • Built and deployed the model using TensorFlow/Keras within an Android application (Java/Kotlin).	2023 – 10/2024

Skills

Programming: C, C++, Python, SQL	CS Fundamentals Data Structures & Algorithms, Operating Systems, OOP, SDLC
ML and DL: — Proficient Supervised Learning, Feature Engineering, Model Evaluation, Ensemble Methods , Train/Validation Pipelines,CNN, RNN, LSTM	Computer Vision: — Proficient Object Detection (YOLO), Image Processing, Feature Extraction, Anomaly Detection
Frameworks & Libraries: — Proficient PyTorch, TensorFlow, Keras, OpenCV, NumPy, Pandas	NLP & LLMs: Text Preprocessing, Embeddings, Sequence Models,Prompt Engineering,LangChain, LangGraph, Generative AI
Databases & Tools: — Proficient MongoDB, SQL, Git	

Internship&Certification

Artificial intelligence course (AI 102) Microsoft Azure AI Engineer Certification — AI 102 (Artificial Intelligence),CSR Technology Initiative — Cognizant Foundation & IECT Academy, Kerala	Web Development Internship — IBM(via Verzeo) Developed and deployed web-based applications using modern development practices.
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Achievements

Placement Coordinator, Core coordinator of Tech Fest of CSE department, Program committee member of college cultural programs and coordinator	2024
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Languages

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